CONSECUTIVE INTEGERS FREE OF CERTAIN PRIME FACTORS

WOUTER VAN DOORN AND QUANYU TANG

Abstract. Let n_k denote the least integer $n > 2k$ such that $(n-k)(n-k+1)\cdots(n-1)$ is not divisible by any prime in the interval $(k, 2k)$. Confirming a conjecture of Erdős, we prove that, for all sufficiently large k,

$$n_k > e^{\frac{\log^2 k}{20 \log\log k}}.$$

1. Introduction

In [6] Erdős writes[1]

> *Several times during my long life, I was led to questions of the following type. Estimate, as well as you can, the size of the smallest integer $n_k > 2k$ for which $\prod_{1\le i\le k}(n-i)$ has no prime factor p satisfying $k < p < 2k$. I would expect that $n_k > k^d$ for every d if $k > k_0(d)$, but that $n_k < e^{\varepsilon k}$ for every $\varepsilon > 0$ if $k > k_0(\varepsilon)$. However, I could prove nothing non-trivial.*

Erdős and Graham repeated this request in [7], and added

> *We can prove $n_k > k^{1+c}$ but no doubt much more is true.*

Estimating n_k is now recorded as Erdős Problem #451 on Bloom's website [3], and no proofs of non-trivial bounds have appeared in the literature, as far as the authors are aware.

For a quick upper bound, we have

$$n_k \le \prod_{\substack{k<p<2k \\ p \text{ prime}}} p = e^{(1+o(1))k}$$

by the prime number theorem. A natural heuristic suggests that the true size of n_k is smaller, however, and is on the scale

$$\exp\left(\Theta\left(\frac{k}{\log k}\right)\right).$$

The details of this heuristic, together with related numerical evidence, are discussed in [11]; computed values of n_k are also recorded in OEIS sequence A386620 [9].

In this paper we focus on lower bounds, and prove Erdős' conjecture that n_k grows super-polynomially fast, by showing that $n_k > e^{\frac{c\log^2 k}{\log\log k}}$ holds for some absolute constant c and all $k > 1$. More precisely, with $\theta \in \left(\frac{2}{5}, \frac{3}{5}\right)$ a constant such that, for all sufficiently large k, the interval $I = I_k := (k, k+k^\theta)$ contains $\gg_\theta \frac{k^\theta}{\log k}$ primes (which exists by e.g. [2]), this paper is then dedicated to proving the following theorem.

Theorem 1.1. *For all sufficiently large integers k and all $n \in \mathbb{N}$ with $2k < n \le e^{\frac{\log^2 k}{20\log\log k}}$, the product $(n-k)\cdots(n-1)$ is divisible by some prime $p \in (k, k+3k^\theta)$.*

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[1]Notation slightly altered to match notation in [7].

For the proof of this lower bound we turn to the similar problem of estimating the smallest $n > k+1$ such that the binomial coefficient $\binom{n}{k}$ has no prime factors smaller than or equal to k. This function was studied by Ecklund, Erdős and Selfridge in [5], where they proved that there exists an absolute constant $c > 0$ such that the bounds $k^{1+c} < n < e^{(1+o(1))k}$ hold for all $k \in \mathbb{N}$. Estimating this quantity is now Erdős Problem #1095 [4], and the best known lower bound of $e^{c\log^2 k}$ (for some absolute constant $c > 0$) was proven by Konyagin [8]. It is Konyagin's proof that we will imitate in the coming sections.

Declaration of AI usage. Even though this paper is completely human-written, the core idea of applying the same arguments used in [8] in order to prove the lower bound $n_k \geq e^{\frac{c\log^2 k}{\log\log k}}$ was conceived of by ChatGPT 5.5 Pro. The original paper that ChatGPT wrote is still available at [10], for transparency's sake. Furthermore, in this same GitHub repository one can also find Lean formalizations of all theorems in this paper, including [8, Theorem 2] that we apply. Apart from the external result on primes in short intervals from [2], these formalizations are fully self-contained. They were obtained by the automated theorem proving tool Aristotle from Harmonic [1].

2. Small n

First assume $2k < n \leq \frac{1}{2}k^{2-\theta}$, which implies that there exists an integer m with $2 \leq m \leq \frac{1}{2}k^{1-\theta}$ such that $mk < n \leq (m+1)k$. Then there are two cases, either

$$mk < n < mk + mk^\theta \qquad \text{or} \qquad mk + mk^\theta \leq n \leq (m+1)k.$$

In the first case, let p be a prime with $p \in \left(k + \frac{m}{m-1}k^\theta, k + \frac{2m-1}{m-1}k^\theta\right)$. We then claim that $(m-1)p$ occurs as a factor in $(n-k)\cdots(n-1)$. Indeed, on the one hand we have

$$(m-1)p > (m-1)k + mk^\theta > n-k,$$

while on the other hand we have

$$(m-1)p < (m-1)k + (2m-1)k^\theta < (m-1)k + k < n.$$

In the second case, let p be a prime with $p \in I$. We then have

$$mp > mk \geq n-k$$

and

$$mp < mk + mk^\theta \leq n,$$

so that in this case mp occurs as a factor in $(n-k)\cdots(n-1)$.

3. Medium n

The medium case is where $\frac{1}{2}k^{2-\theta} < n \leq \frac{k^2}{\log^2 k}$. With $\|x\|$ the distance from a real number x to the nearest integer, let K be the number of integers $m \in I$ such that $\|\frac{n}{m}\| < \frac{1}{k^{1-\theta}}$. We then claim that it suffices to show $K = o\left(\frac{k^\theta}{\log k}\right)$. To see why this is indeed sufficient, we note that it implies that there exists a prime $p \in I$ with $\|\frac{n}{p}\| \geq \frac{1}{k^{1-\theta}}$, by definition of θ. And this in turn implies that reducing n modulo p gives

$$n \bmod p \in [k^\theta, p - k^\theta] \subset [1, k],$$

so that p divides $(n-k)\cdots(n-1)$.

For all $m \in I$, we have $\frac{n}{m} \in \left(\frac{n}{k+k^\theta}, \frac{n}{k}\right) \subseteq J := \left(\left\lfloor \frac{n}{k+k^\theta} \right\rfloor, \left\lceil \frac{n}{k} \right\rceil\right)$ with

$$|J \cap \mathbb{Z}| \leq 3 + \frac{n}{k} - \frac{n}{k+k^\theta} = 3 + \frac{nk^\theta}{k(k+k^\theta)} < 3 + \frac{n}{k^{2-\theta}}.$$

For every integer $h \in J$, if $\frac{n}{m} \in \left(h - \frac{1}{k^{1-\theta}}, h + \frac{1}{k^{1-\theta}}\right)$, then

$$\left|m - \frac{n}{h}\right| = \frac{m}{h}\left|\frac{n}{m} - h\right| < \frac{2k}{\frac{n}{k}} \frac{1}{k^{1-\theta}} = \frac{2k^{1+\theta}}{n}.$$

This implies that for every integer $h \in J$ there are at most $1 + \frac{4k^{1+\theta}}{n}$ values of m such that $\left|\frac{n}{m} - h\right| < \frac{1}{k^{1-\theta}}$, which shows

$$\begin{aligned} K &< \left(1 + \frac{4k^{1+\theta}}{n}\right)|J \cap \mathbb{Z}| \\ &< 3 + \frac{12k^{1+\theta}}{n} + \frac{n}{k^{2-\theta}} + 4k^{2\theta-1} \\ &< 3 + 28k^{2\theta-1} + \frac{\frac{k^2}{\log^2 k}}{k^{2-\theta}} \\ &= o\left(\frac{k^\theta}{\log k}\right). \end{aligned}$$

4. Intermezzo on a theorem by Konyagin

To show that $K = o\left(\frac{k^\theta}{\log k}\right)$ also holds for all n with $\frac{k^2}{\log^2 k} < n \leq e^{\frac{\log^2 k}{20 \log\log k}}$ we will make use of the following general upper bound on K.

Theorem 4.1. *Let r be an integer with $2 \leq r \leq \frac{1}{2}k^{1-\theta}$ and let $\lambda \geq 1$ be arbitrary. We then have*

$$K \ll k^\theta \left(\left(\frac{nr!\lambda^r}{k^{r+1}}\right)^{\frac{1}{2r-1}} + \left(\frac{k^{r+\theta}}{nr!\lambda^r}\right)^{\frac{1}{r-1}} + \left(\frac{(r+1)\lambda}{k}\right)^{\frac{1}{2r}}\right) + r\lambda. \qquad (1)$$

Proof. We apply [8, Theorem 2], with

$$\begin{aligned} N &:= k^\theta, \\ W &:= 1, \\ f(x) &:= \frac{(-1)^r n}{k+x}, \\ D_r &:= \frac{nr!}{k^{r+1}}, \\ \delta &:= \frac{1}{k^{1-\theta}}. \end{aligned}$$

The only non-trivial conditions one has to check are the required bounds in [8] on the iterated derivatives of f. But as $f^{(i)}(x) = \frac{(-1)^{r+i}ni!}{(k+x)^{i+1}}$, the upper bounds $f^{(r)}(x) \leq D_r$ and $|f^{(r+1)}(x)| \leq D_{r+1}$ quickly follow. To see why $f^{(r)}(x) \geq \frac{1}{2}D_r$ also holds, we use the assumption

$r \leq \frac{1}{2}k^{1-\theta}$, as follows.

$$\begin{aligned} f^{(r)}(x) &= \frac{nr!}{(k+x)^{r+1}} \\ &\geq \frac{nr!}{\left(k\left(1+\frac{1}{k^{1-\theta}}\right)\right)^{r+1}} \\ &= \frac{D_r}{\left(1+\frac{1}{k^{1-\theta}}\right)^{r+1}} \\ &> \frac{D_r}{\left(1+\frac{1}{2r}\right)^{r+1}} \\ &> \frac{1}{2}D_r. \end{aligned}$$

□

5. Medium-large n

The medium-large case is where $\frac{k^2}{\log^2 k} < n \leq \frac{1}{2}k^{2+\theta}$. We then apply Theorem 4.1 with $\lambda := \sqrt{\frac{k^{2+\theta}}{2n}}\log k$ and $r := 2$, which gives

$$\begin{aligned} K &\ll k^{\theta}\left(\left(\frac{nr!\lambda^r}{k^{r+1}}\right)^{\frac{1}{2r-1}} + \left(\frac{k^{r+\theta}}{nr!\lambda^r}\right)^{\frac{1}{r-1}} + \left(\frac{(r+1)\lambda}{k}\right)^{\frac{1}{2r}}\right) + r\lambda \\ &= k^{\theta}\left(k^{\frac{\theta-1}{3}}(\log k)^{\frac{2}{3}} + \frac{1}{\log^2 k} + \left(\frac{(r+1)\lambda}{k}\right)^{\frac{1}{2r}}\right) + r\lambda \\ &\ll k^{\theta}\left(k^{\frac{\theta-1}{3}}(\log k)^{\frac{2}{3}} + \frac{1}{\log^2 k} + k^{\frac{\theta-2}{8}}\sqrt{\log k}\right) + k^{\frac{\theta}{2}}\log^2 k \\ &= o\left(\frac{k^{\theta}}{\log k}\right). \end{aligned}$$

6. Large n

Finally, we deal with the case of large n. More precisely, let us take $c := \frac{1}{20}$ and assume $\frac{1}{2}k^{2+\theta} < n \leq e^{\frac{c\log^2 k}{\log\log k}}$. With r defined as the smallest positive integer such that $nr! \leq k^{r+\theta}$, we have $r \geq 3$ by the assumption on n. Furthermore, in general we see that such an r does indeed exist and $r \leq \frac{1}{2}k^{1-\theta}$, because with $r_0 := \left\lceil \frac{2c\log k}{\log\log k} \right\rceil$ we have $r_0 \leq \frac{1}{2}k^{1-\theta}$ and

$$\begin{aligned} k^{r_0+\theta} &> k^{\frac{2c\log k}{\log\log k}} \\ &\geq ne^{\frac{c\log^2 k}{\log\log k}} \\ &> ne^{r_0\log r_0} \\ &> nr_0!. \end{aligned}$$

We further define $\lambda := \left(\frac{k^{r+1-(1-\theta)\frac{2r-1}{3r-2}}}{nr!}\right)^{\frac{1}{r}}$, which is larger than 1 by definition of r. Hence, Theorem 4.1 applies, and with this definition of λ, the first two terms within the brackets in

Equation (1) are both equal to $k^{\frac{\theta-1}{3r-2}}$. Since $3r-2 < \frac{7c\log k}{\log\log k}$ and $\theta < \frac{3}{5}$, this is smaller than

$$k^{\frac{(\theta-1)\log\log k}{7c\log k}} = (\log k)^{\frac{\theta-1}{7c}} = o\left(\frac{1}{\log k}\right).$$

As for the third term, thanks to the minimality of r we know $n(r-1)! > k^{r-1+\theta}$. This gives

$$\lambda = \left(\frac{k^{r+1-(1-\theta)\frac{2r-1}{3r-2}}}{nr!}\right)^{\frac{1}{r}} < k^{\frac{2-\theta-(1-\theta)\frac{2r-1}{3r-2}}{r}} < k^{\frac{2-\theta}{r}}.$$

Hence, as $r+1 < k^{\frac{\theta}{r}}$ if k is large enough, the third term is bounded by

$$\left(\frac{(r+1)\lambda}{k}\right)^{\frac{1}{2r}} < k^{\frac{1}{r^2}-\frac{1}{2r}} \le k^{\frac{-1}{6r}} < (\log k)^{\frac{-1}{13c}} = o\left(\frac{1}{\log k}\right).$$

For the final additive term of λr we also apply the minimality of r, which implies

$$\lambda r < k^{\frac{2-\theta-(1-\theta)\frac{2r-1}{3r-2}}{r}} r \le k^{\frac{9-2\theta}{21}} r_0 = o\left(\frac{k^\theta}{\log k}\right),$$

by the assumption $\theta > \frac{2}{5} > \frac{9}{23}$ and the fact that $\frac{2-\theta-(1-\theta)\frac{2r-1}{3r-2}}{r}$ as a function over $r \in [3, r_0]$ is maximized at $r = 3$. As the full range of integers n with $2k < n \le e^{\frac{\log^2 k}{20\log\log k}}$ is now dealt with, this finishes the proof of Theorem 1.1.

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Groningen, the Netherlands
Email address: wonterman1@hotmail.com

School of Mathematics and Statistics, Xi'an Jiaotong University, Xi'an 710049, P. R. China
Email address: tangquanyu827@gmail.com